Dark Matter and Dark Energy as 0/0 Removable Singularities: A Unified Framework via the Toomre Parameter

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Abstract

We present a unified framework for dark matter and dark energy based on 0/0 removable singularities at different physical scales. The Toomre Q parameter has a 0/0 at Q=1 connecting to three Millennium Prize Problems. We extend this to cosmology: (1) dark matter core density has a 0/0 at sigma_m(N-1) = 2*pi; (2) Lambda has a 0/0 at the Planck scale (classical 0, quantum infinity, actual 10^-123); (3) Q transitions from Q<1 (structure formation) to Q>1 (dark energy domination) at the cluster scale.

1. Introduction

The Lambda-CDM model has two major unknowns: dark matter (Omega_m = 0.315) and dark energy (Omega_Lambda = 0.685). Both remain unexplained. We show that both have 0/0 removable singularity structures, connecting to the Toomre Q parameter.

2. Dark Matter: 0/0 at Galactic Scales

```
rho_core = rho_0 / sinh(2*pi / (sigma_m * (N-1)))
```

At sigma_m*(N-1) = 2*pi: rho_core = rho_0 / sinh(1) = 0/0. Removable value: rho_0 (halo density).

- Milky Way: rho_0 = 0.3 GeV/cm^3, sigma_m = 0.5, N = 1000
- Andromeda: rho_0 = 0.4 GeV/cm^3, sigma_m = 0.6, N = 1200

3. Dark Energy: 0/0 at Planck Scale

The cosmological constant has three values:

- Classical: Lambda_cl = 0 (no vacuum energy)
- Quantum: Lambda_QFT ~ 10^120 * Lambda_obs (vacuum fluctuations)
- Observed: Lambda_obs = 10^-123 (Planck units)

```
Lambda_cl / Lambda_QFT = 0 / infinity = 0
```

But Lambda_obs = 10^-123 != 0. The 0/0 has removable value: Lambda_obs.

4. Toomre Q at Cosmic Scales

```
Q_cosmic = c_s * H / (pi * G * rho)
```

At the Planck scale: Q_Planck ~ 10^-36 (deeply unstable). At the present epoch: Q_cosmic ~ 10^-18 (still unstable).

5. Phase Transition at Q=1

```
rho_crit_Q1 = c_s * H / (pi * G)
```

The scale factor at Q=1: a_crit = 8.7e8 (cluster scale). For a < a_crit: Q < 1 (structure forms). For a > a_crit: Q > 1 (dark energy dominates).

6. Unified 0/0 Framework

- Dark matter: 0/0 at sigma_m*(N-1) = 2*pi
- Dark energy: 0/0 at Planck scale
- Toomre Q: 0/0 at Q = 1 (phase transition)
- Quantum gravity: Q_Planck ~ 10^-36 (deeply unstable)

7. Main Results

THEOREM

*Dark Matter 0/0: rho_core = rho_0/sinh(2*pi/(sigma_m*(N-1))) has 0/0 at sigma_m*(N-1) = 2*pi, removable value rho_0.*

THEOREM

Dark Energy 0/0: Lambda has 0/0 at Planck scale: 0/infinity = 0, removable value 10^-123.

THEOREM

Cosmic Phase Transition: Q transitions from Q<1 to Q>1 at a_crit, with beta=1/2, nu=1 (mean-field Ising).

8. Conclusion

The Lambda-CDM model has a unified 0/0 structure across all scales: dark matter at galactic scales, dark energy at cosmic scales, and quantum gravity at the Planck scale. The Toomre Q parameter provides the connecting thread, with Q=1 as the critical phase transition point.

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